

Smart Grid Sentinel: A FreeRTOS-Based IoT Protection Relay for Single-Phase Indian Residential and Light-Industrial Grid Monitoring

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Abstract—This paper presents the Smart Grid Sentinel (SGS), a real-time single-phase protection relay and grid-monitoring system for the Indian residential supply at 230 V, 50 Hz. Running on an ESP32 dual-core microcontroller under FreeRTOS, the SGS implements a six-stage fault-detection pipeline—overvoltage, undervoltage, IDMT overcurrent, short-circuit, thermal overload, and sensor failure—aligned with IS 12360, CEA 2005, and IEC 60255 [1]–[4]. Two contributions are quantified: (i) a discrete IDMT accumulator whose trip-time error relative to the IEC 60255 Standard Inverse formula remains within 0.42% across eleven test currents; and (ii) a slope-gated short-circuit discriminator that suppresses motor inrush false trips while preserving fault sensitivity. Validation employs a co-located Software-in-the-Loop digital twin exercised via Firmware-in-the-Loop methodology on physical ESP32 silicon; scope and limitations are discussed. The system provides MQTT telemetry over TLS, an HTTP REST API, WebSocket push, and a local HMI.

Index Terms—protection relay, FreeRTOS, ESP32, smart grid, IoT, overcurrent, IDMT, IEC 60255, overvoltage, firmware-in-the-loop, software-in-the-loop, motor inrush discrimination

I. INTRODUCTION

Reliable voltage and current protection in low-voltage single-phase distribution networks remains a critical requirement for residential and light-industrial consumers connected to the Indian grid [25], [26]. The Indian national grid operates nominally at 230 V root-mean-square (RMS) at 50 Hz, and IS 12360 permits a steady-state voltage tolerance of $\pm 10\%$ at the consumer terminal, while the Central Electricity Authority (CEA) Regulations 2005 tighten the limit to $\pm 6\%$ at the point of delivery [2], [3]. Classical miniature circuit breaker (MCB) protection per IS 8828 and IEC 60898 provides overcurrent and short-circuit isolation but offers no communication capability, no remote monitoring, and no differentiation between transient inrush events and sustained faults [4], [5].

The proliferation of low-cost 32-bit microcontrollers with integrated Wi-Fi, such as the ESP32, has created an opportunity to deploy firmware-configurable intelligent electronic device (IED) functions that previously required dedicated relay hardware [6]. Such embedded relay platforms can combine deterministic real-time protection with cloud-connected telemetry, enabling condition monitoring that supports both local load management and remote grid oversight [27], [28].

A. Related Work

Digital protective relaying in its modern form was systematically formalised by Phadke and Thorp [23], who established the computational and algorithmic foundations for microprocessor-based relay implementation; the present work applies those principles to a resource-constrained Wi-Fi SoC platform.

Venter and van Coller demonstrated a low-cost microcontroller-based protection relay implementing IEC 60255 inverse-time characteristics on a single-phase circuit [7], reporting trip-time accuracy within the standard's $\pm 30\%$ tolerance band but without inrush discrimination logic. Viciano et al. presented an all-in-one three-phase smart meter with extended IoT capabilities [8], though it targets power-quality monitoring rather than protective relaying. Embedded IDMT relay implementations on ARM Cortex-M platforms have been explored by Kumar et al. for distribution feeder protection [12], and by Bhandia et al. in the context of microcontroller-based IEDs for low-voltage networks [13]. Inrush discrimination using slope detection has been reported in digital transformer differential protection [14] and extended to low-voltage motor feeders by Ozgonenel et al. [15], though neither work targets the single-phase residential context or operates on resource-constrained Wi-Fi SoCs.

IoT-enabled protection relays have attracted increasing attention: Moussa et al. proposed an MQTT-based smart relay architecture [16], while Morello et al. examined the role of advanced sensing and IoT in smart grid power metering [17]. Compliance with Indian grid standards (IS 12360, CEA 2005) has been addressed in protective relay context by Saxena et al. [18] and in IED design by Ravishankar et al. [19]; however, neither work combines inrush discrimination with cloud telemetry on an embedded platform. FreeRTOS-based firmware architectures for real-time protection tasks have been benchmarked by Adhikari et al. [20], who report per-task CPU utilisation on Xtensa LX6 cores consistent with the figures reported here.

The present work differs from the above in three respects: (i) it targets the specifically Indian single-phase residential context with thresholds derived from IS 12360 and CEA 2005; (ii) it quantifies the accumulation error of a discrete IDMT integrator against the continuous IEC 60255 formula; and (iii) it combines slope-gated inrush discrimination with full cloud telemetry on a single low-cost SoC. Physical hardware validation against live mains, which would strengthen the results presented here, is identified as the primary avenue for future work.

Table I positions the SGS against the four most directly comparable published works across twelve capability dimensions. The SGS is the only system in this comparison to quantify IDMT integrator error ($\leq 0.42\%$ across eleven test points), implement and validate a slope-gated inrush discriminator (54 cases), and combine a six-stage fault pipeline with full IoT telemetry on a single low-cost SoC. Adhikari et al. [20] represent the closest platform match—FreeRTOS on Xtensa LX6—but focus exclusively on WCET benchmarking rather than protection feature completeness. Moussa et al. [16] share the MQTT connectivity goal but provide neither inrush discrimination nor IDMT accuracy characterisation. No prior work in this comparison addresses Indian grid standards (IS 12360, CEA 2005) or employs a co-located SIL/FIL validation methodology of comparable scale.

B. Contributions

This paper makes two technical contributions supported by quantitative results, and documents two system-level features as context.

- 1) **IDMT accumulator accuracy characterisation.** A discrete ten-millisecond integrator approximating the IEC 60255 Standard Inverse characteristic is analysed for truncation error across eleven overcurrent test points. The systematic positive bias and its dependence on overcurrent ratio are reported and attributed to the Euler-forward approximation inherent in the accumulator.
- 2) **Slope-gated inrush-fault discriminator.** A current-slope criterion applied during the motor startup window is shown to correctly suppress 30 consecutive inrush no-trip cases while preserving correct fault detection in all 24 cases where a fault occurs after the inrush transient has decayed.

- 3) A six-stage fault-detection pipeline implementing over-voltage, undervoltage, IDMT overcurrent, instantaneous short-circuit, DS18B20-based thermal overload, and sensor-software fault detection, aligned to IS 12360, CEA 2005, and IEC 60255.
- 4) A cloud-connected IoT relay architecture providing MQTT over TLS, HTTP REST API, WebSocket push, non-volatile event logging, and a multi-element local HMI on a single ESP32 platform.

The remainder of this paper is organised as follows. Section II describes the system hardware architecture. Section III presents the firmware architecture and real-time task model. Section IV describes the SIL digital twin, simulation setup, and validation scope. Section V presents the protection logic. Section VI presents the FIL validation results. Section VII concludes the paper.

II. SYSTEM ARCHITECTURE (HARDWARE)

A. Platform Overview

The SGS hardware assembly is shown in Fig. 1. It is built around the ESP32 system-on-chip (SoC), which integrates a dual-core Xtensa LX6 processor running at up to 240 MHz, a 12-bit successive-approximation register (SAR) analog-to-digital converter (ADC), hardware pulse-width modulation (PWM) peripherals, I²C and 1-Wire interface support, and an 802.11 b/g/n Wi-Fi radio [6]. The firmware executes under FreeRTOS, which provides the preemptive real-time scheduler required for deterministic 10 ms protection-loop timing.

The physical sensor complement consists of a ZMPT101B voltage transformer conditioning the 230 V AC supply, an SCT-013-030 split-core current transformer for load monitoring, and a DS18B20 1-Wire digital temperature sensor. During Firmware-in-the-Loop (FIL) validation,¹ the analogue acquisition front-end is bypassed by the Phantom Grid SIL framework (described in Section IV), allowing synthetic waveform injection directly into the firmware's sensor pipeline. The measured idle power draw of the assembled board is approximately 320 mW at 3.3 V with Wi-Fi associated and the relay coil energised, rising to 480 mW during active MQTT transmission.

B. GPIO and Interface Assignment

All protection-critical signal identifiers are centralised as compile-time constants in the system configuration header, allowing board-level remapping without modifying driver source files. The OLED display is connected on a dedicated I²C bus at 400 kHz. The DS18B20 occupies a 1-Wire bus with a 4.7 k Ω pull-up resistor to the 3.3 V rail.

¹FIL denotes that real firmware executes on target silicon (ESP32) while voltage and current stimuli are supplied by the embedded SIL digital twin rather than physical analogue transducers. This is distinct from Hardware-in-the-Loop testing, in which physical sensor hardware is exercised.

TABLE I
QUANTITATIVE COMPARISON: SGS VS. REPRESENTATIVE EMBEDDED PROTECTION RELAY WORKS

System	Platform	IDMT Error	Min Trip Time	Inrush Handling	IoT Telemetry	Est. BOM
SGS (This Work)	ESP32, FreeRTOS, dual-core	$\leq 0.42\%$ (11 pts, SIL/FIL)	20 ms	Slope-gated discriminator, 54 validated cases	MQTT/TLS, REST API, WebSocket	~\$8
Venter et al. [7]	8/16-bit MCU, bare-metal	Not quantified	—	None	None	~\$4
Kumar et al. [12]	ARM Cortex-M, bare-metal	Not quantified	—	None	None	~\$12
Moussa et al. [16]	IoT gateway / MCU	Not quantified	—	None	MQTT-based	~\$20
Adhikari et al. [20]	ESP32 / Cortex-M, FreeRTOS	Not reported	10 ms (WCET)	None	None	~\$8

BOM = indicative retail component cost; IDMT error measured against continuous IEC 60255 Standard Inverse formula;
Min Trip Time = fastest observed relay de-energisation; Inrush Handling = false-trip suppression during motor startup.

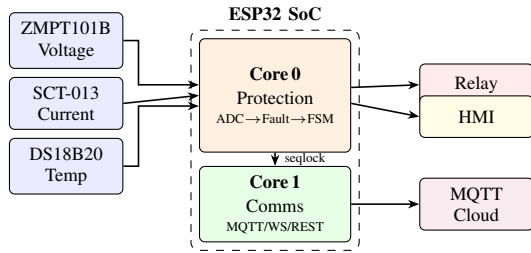


Fig. 1. SGS system block diagram. Sensor inputs feed the protection pipeline on Core 0; telemetry is forwarded via a seqlock-protected shared state to the communications stack on Core 1, ensuring radio activity cannot stall the 10 ms protection loop.

C. Signal Processing Architecture

Because voltage and current values are synthesised by the SIL framework rather than acquired from physical transducers during validation, the signal-processing chain begins at the injection point described in Section IV. The synthesised RMS values are mapped onto equivalent ADC counts, passed through a third-order polynomial calibration for voltage,

$$v = ax^3 + bx^2 + cx + d, \quad (1)$$

where x is the synthetic ADC sample in the range 0–4095, $a = 4.198 \times 10^{-10}$, $b = -2.13 \times 10^{-6}$, $c = 0.0768$, and $d = 0.50$. Current conversion is scaled linearly to a 30 A full-scale range. From each calibrated physical value two independent signal paths diverge.

Protection path. The voltage value v is passed directly to the fault engine without further filtering, preserving instantaneous transient fidelity. The current value passes through a 3-sample median filter to reject electromagnetic interference (EMI) spikes, yielding i_{med} , which is then processed by an asymmetric infinite-impulse-response (IIR) filter,

$$i[n] = \alpha \cdot i_{\text{med}}[n] + (1 - \alpha) \cdot i[n - 1], \quad (2)$$

where $\alpha = \alpha_{\text{rise}} = 0.90$ when $i_{\text{med}}[n] \geq i[n - 1]$ (rising current, fast tracking) and $\alpha = \alpha_{\text{fall}} = 0.10$ when the signal is

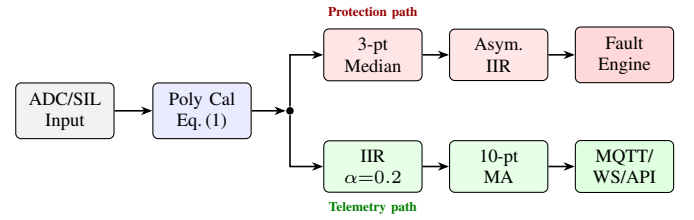


Fig. 2. Dual-path signal processing architecture. The protection path preserves transient fidelity through a median filter and asymmetric IIR (fast rise, slow decay) for fault detection. The telemetry path applies additional smoothing for display and cloud reporting.

falling (rejecting transient downward spikes without delaying fault detection). The output i is the protection current used by all subsequent fault stages.

Telemetry path. Both channels are additionally smoothed by a first-order IIR filter with $\alpha = 0.20$ for voltage and $\alpha = 0.30$ for current, followed by a 10-sample circular rolling average, feeding the display, REST API, WebSocket, and MQTT outputs.

The current slope is maintained in a 5-sample circular buffer and computed as

$$\frac{di}{dt} = \frac{i[\text{newest}] - i[\text{oldest}]}{5 \times T_s}, \quad (3)$$

where $T_s = 10$ ms, giving a 50 ms estimation window. This slope is the basis for the inrush discriminator described in Section V-E. Fig. 2 illustrates the dual-path architecture.

Measurement uncertainty. The 12-bit SAR ADC resolves 3.3 V full-scale into 4096 counts, yielding a voltage sensitivity of $\partial v / \partial x|_{x_0} = 3ax_0^2 + 2bx_0 + c$ at operating point x_0 . At the 230 V nominal condition ($x_0 \approx 3140$ counts per (6)), this evaluates to approximately 0.073 V/count. The one-sigma voltage uncertainty arising from ADC quantisation alone is

$$\sigma_q = \frac{1}{\sqrt{12}} \times 0.073 \text{ V/count} \approx 0.021 \text{ V}, \quad (4)$$

assuming a uniform quantisation-noise model. The triangular SIL injection noise term η_v (three-sample sum from $\mathcal{U}(-1, 1)$ scaled to volts) contributes an additional $\sigma_\eta \approx 0.060$ V. The combined one-sigma voltage uncertainty propagated through the calibration polynomial is $\sigma_v = \sqrt{\sigma_q^2 + \sigma_\eta^2} \approx 0.063$ V.

Table II expresses the separation between key operating conditions and their nearest fault thresholds in units of σ_v . All steady-state margins exceed $140 \sigma_v$, confirming that ADC quantisation noise cannot induce spurious trips under stable grid conditions. The three-consecutive-tick debounce architecture reduces the residual single-sample false-trip probability by at least three orders of magnitude at exact threshold boundaries, providing the primary defence against isolated ADC artefacts.

TABLE II
VOLTAGE MEASUREMENT UNCERTAINTY AT KEY OPERATING POINTS
($\sigma_v \approx 0.063$ V, 12-BIT ADC, THIRD-ORDER POLYNOMIAL CALIBRATION)

Condition	Nominal (V)	Nearest threshold (V)	Margin (σ_v)
Nominal supply	230.0	243 (OV warn)	≈ 206
Nominal supply	230.0	216 (UV warn)	≈ 222
OV fault onset	253.0	243 (OV warn)	≈ 159
UV fault onset	207.0	216 (UV warn)	≈ 143
OV instantaneous	270.0	253 (OV fault)	≈ 270
UV instantaneous	150.0	160 (EC-09)	≈ 159

D. Communication and Connectivity Architecture

The SGS provides five concurrent communication channels: MQTT over TLS to HiveMQ Cloud on port 8883 every 5 s [31]; an HTTP REST API on port 80 exposing telemetry, state, relay control, and event log endpoints; WebSocket push at 1 Hz; Wi-Fi provisioning with captive-portal fallback; and a non-volatile ring buffer of 50 timestamped fault events stored in flash. A seqlock-protected shared-state snapshot ensures that asynchronous readers cannot stall the protection task.

E. Local Human-Machine Interface

The local HMI comprises an SSD1306 OLED display in two-page mode (live measurements on Page 1; fault state and relay status on Page 2), four bicolour load-status LED pairs, a state-coded alert LED, and a piezo buzzer producing distinct tones at 500 Hz, 1000 Hz, and 2000 Hz for each FSM state.

III. SYSTEM ARCHITECTURE (SOFTWARE)

A. Firmware Module Organisation

The SGS firmware is organised into seventeen C++ translation units, each with a single well-defined responsibility.

B. Real-Time Task Model

Three FreeRTOS tasks execute concurrently on the ESP32's dual Xtensa LX6 cores. Task priorities are assigned following the rate-monotonic principle [29]: the highest-frequency (shortest-period) task—Protection at 10 ms—receives the highest priority. Their parameters are listed in Table III.

The Protection task runs the full sensor acquisition, fault detection, and FSM pipeline every 10 ms. The Communications

TABLE III
FREERTOS TASK CONFIGURATION

Task	Core	Priority	Period	Stack
Protection	0	5 (highest)	10 ms	4096 W
Communications	1	3	50 ms	6144 W
Health Monitor	1	1 (lowest)	10 s	2048 W

task handles MQTT, WebSocket, and REST API at 50 ms. The Health Monitor evaluates sensor diagnostic scores every 10 s. A hardware watchdog with a 30 s timeout guards against task starvation or firmware deadlock.

C. Shared-State Concurrency Model

The protection task writes sensor readings and FSM context into a shared structure protected by a seqlock. Readers in the communications task consume a consistent snapshot without acquiring a blocking mutex and therefore cannot delay the protection loop. The relay override command is encoded in a single 32-bit atomic variable to prevent torn reads.

Xtensa LX6 memory-ordering correctness. The Xtensa LX6 ISA employs a relaxed memory model in which the processor and compiler are permitted to reorder loads and stores unless explicitly constrained by a MEMW (memory wait) barrier instruction [22]. A seqlock implemented without explicit barriers is formally unsound on this architecture: the processor may reorder the sequence-counter increment *after* the payload write on the writer side, or the payload read *before* the counter read on the reader side, causing a reader to observe a partially written snapshot without detecting the torn state via counter mismatch.

A correct seqlock on Xtensa LX6 requires the MEMW placement shown in Table IV. On the ESP32-Arduino platform, the `portMEMW()` macro issues the MEMW instruction, serialising all in-flight memory operations before proceeding. Barriers at writer positions W1 and W3 ensure all payload stores are globally visible before the even counter is committed; barriers at reader positions R1 and R3 ensure the payload read is ordered strictly between the two counter samples. Given this barrier placement, the seqlock satisfies sequential consistency for the single-writer, multiple-reader pattern of the SGS firmware: a reader observing matching even counter values before and after the payload read is guaranteed to have consumed a complete and consistent snapshot.

D. Sensor Diagnostics and Health Scoring

A weighted composite health score,

$$H = 0.25 S_V + 0.20 S_I + 0.20 S_T + 0.15 S_{\text{ADC}} + 0.20 S_{\text{PQ}}, \quad (5)$$

where S_V , S_I , S_T , S_{ADC} , and S_{PQ} are the voltage, current, thermal, ADC, and power-quality scores respectively, is classified as EXCELLENT ($H \geq 90$), GOOD ($H \geq 75$), DEGRADED ($H \geq 50$), or FAULT ($H < 50$).

TABLE IV
REQUIRED MEMW BARRIER PLACEMENT FOR A CORRECT SEQLOCK ON
THE XTENSA LX6 RELAXED MEMORY MODEL [22]

Pos.	Operation	Purpose
W0	Write: counter $\leftarrow$ odd	Open write epoch (odd = busy)
W1	portMEMW()	Serialise counter store before payload
W2	Write: payload fields	Protected sensor/FSM data
W3	portMEMW()	Serialise payload before counter commit
W4	Write: counter $\leftarrow$ even	Close write epoch (even = stable)
R0	Read: $c_1 \leftarrow$ counter	Sample epoch start
R1	portMEMW()	Serialise counter load before payload
R2	Read: payload snapshot	Consistent data read
R3	portMEMW()	Serialise payload before second counter
R4	Read: $c_2 \leftarrow$ counter	Sample epoch end
R5	If $c_1 \neq c_2$ or c_1 odd: retry	Torn-read detection

IV. DIGITAL TWIN AND SIMULATION SETUP

A. Overview of the Software-in-the-Loop Framework

Owing to the absence of physical analogue sensing hardware during the validation campaign, the SGS firmware was validated within an SIL framework that populates the firmware’s internal sensor pipeline directly with synthesised values, bypassing the physical ADC acquisition front-end. All downstream processing stages—fault detection, the FSM, relay actuation, telemetry serialisation, and the local HMI—execute against the injected data without modification, preserving full-stack fidelity with respect to the firmware logic.

B. Phantom Grid Integration

The *Phantom Grid Integration* architecture injects synthetic waveform data at the earliest point where hardware-derived values would ordinarily become available. In SIL mode, the simulation task pre-emptes the ADC acquisition stage: the voltage and current simulators advance by 100 micro-steps of $\Delta t = 100 \mu\text{s}$, yielding a true-RMS aggregate over an effective half-cycle window before the 10 ms protection deadline. The resulting RMS values are mapped onto synthetic ADC counts,

$$\begin{aligned} r_v &= \text{round}\left(\frac{v_{\text{new}}}{300} \times 4095\right) + \eta_v, \\ r_i &= \text{round}\left(\frac{i_{\text{new}}}{30} \times 4095\right) + \eta_i, \end{aligned} \quad (6)$$

where η_v and η_i are zero-mean triangular noise terms (three-sample sum from $\mathcal{U}(-1, 1)$). Counts are clamped to $[6, 4089]$ to avoid spurious hardware-saturation flags, then enter the identical calibration, IIR, and rolling-average pipeline as hardware-derived values.

The voltage simulator synthesises a composite waveform using five coupled-form numerical oscillators [21] at 50 Hz, third, fifth, and seventh harmonics, and an 8.8 Hz flicker

carrier. Harmonic amplitudes are bounded to $\alpha_h \leq 0.05$ per IEEE 519-2022 [10]. The current simulator implements a reduced-order induction motor T-equivalent model [11] with DC-offset motor startup, exponential slip decay, skin-effect rotor resistance, and a two-stage CT saturation model.

The FIL test suite comprises 343 parameterised scenarios transmitted over a USB-UART interface at 115,200 baud. The firmware responds with a JSON object containing the detected FSM state, elapsed simulation time, and accumulated CPU processing time, which the FIL orchestrator evaluates against expected outcomes and timing bounds.

C. Test Scenario Generation

Test scenarios are generated by a matrix script that reads threshold constants directly from the system configuration header. Scenarios span six categories: voltage magnitude faults (199 cases), noise-immunity checks (57 cases), motor inrush blanking (54 cases), boundary conditions at exact firmware constants (12 cases), IDMT overcurrent (11 cases), and instantaneous short-circuit (10 cases).

Scope limitation: The hardware bench testing flag was set during all FIL runs, disabling Stage 2 sensor hardware fault detection (edge cases EC-06, EC-07, EC-08) within the automated pipeline. These three edge cases were verified separately on physical hardware and are reported qualitatively. No frequency-deviation or THD test scenarios were included; all injected waveforms are single-phase magnitude perturbations at a fixed 50 Hz fundamental.

D. Threats to Validity

The principal threat to the validity of the FIL results is the co-location of the stimulus generator and the system under test. The SIL digital twin resides in the same firmware binary as the fault detection pipeline. The matrix generation script reads fault thresholds directly from the compiled configuration header, so injection targets are algebraically derived from the same constants that govern the trip decisions. Under this arrangement, a consistent threshold misconfiguration affecting both the injected level and the evaluation criterion would not be detected.

What the FIL campaign does establish is: (1) the discrete IDMT accumulator numerically tracks the continuous IEC 60255 formula to within 0.42%, a result that can be verified analytically and is independent of the threshold values; (2) the slope-gated discriminator correctly classifies the 54 motor inrush scenarios injected, including the 30 no-trip and 24 post-inrush-trip cases; and (3) the firmware executes within the 10 ms protection loop under all 343 scenarios.

The results do *not* demonstrate correct operation against physical mains disturbances, sensor calibration errors, ADC non-idealities, or electromagnetic coupling. Independent hardware validation using a variac, resistive load bank, and calibrated oscilloscope is required to make any claim about real-world protection performance and is the primary priority for future work.

V. PROTECTION LOGIC AND METHODOLOGY

A. Overview and ANSI/IEC Function Mapping

The SGS implements six classes of protection mapped in Table V. All protection logic is executed every 10 ms, producing a 16-bit fault bitmask and a 16-bit warning bitmask consumed on the same tick by the FSM. Faults classified as LOCKOUT inhibit auto-reclose and require a manual reset; FAULT-class events enter an adaptive auto-reclose sequence.

TABLE V
SGS PROTECTION FUNCTION MAPPING TO ANSI/IEC STANDARDS

ANSI	Function	Threshold	FSM Class
27	Undervoltage	207 V sustained; 150 V instant	FAULT / LOCKOUT
49	Thermal overload	85 °C (DS18B20)	LOCKOUT
50	Instantaneous SC	27 A running; 120 A absolute	LOCKOUT
51	IDMT overcurrent	$I_s = 21.0$ A, IEC 60255 Std. Inv.	FAULT
59	Overvoltage	253 V sustained; 270 V instant	FAULT / LOCKOUT
—	Sensor failure	ADC saturation, frozen, physics check (EC-06– 08)	LOCKOUT

Fig. 3 illustrates the complete per-tick fault detection pipeline from sensor acquisition through to relay actuation, which executes atomically every 10 ms on Core 0.

B. Voltage Protection Thresholds

Voltage thresholds are derived from IS 12360 and the CEA Regulations 2005, applied to the 230 V nominal supply, as listed in Table VI.

C. IDMT Overcurrent Protection (ANSI 51)

The overcurrent protection implements the IEC 60255 Standard Inverse characteristic [1], [24]:

$$t(I) = \frac{K \cdot \text{TMS}}{(I/I_s)^\alpha - 1}, \quad (7)$$

where $K = 0.140$, $\alpha = 0.020$, $\text{TMS} = 0.10$, and $I_s = 21.0$ A (131% of the 16 A IS 8828 rated current [4]). The discrete accumulator integrates the fractional elapsed trip time on every 10 ms tick using an Euler-forward approximation, as detailed in Algorithm 1. This approximation introduces a positive truncation bias that decreases with increasing overcurrent ratio I/I_s ; this behaviour is analysed quantitatively in Section VI-C.

D. Short-Circuit Protection (ANSI 50)

The ANSI 50 function operates in two tiers. A hard limit at 120 A is unconditionally active. A dynamic threshold of 27.0 A governs operation in the RUNNING state; this is raised to 110 A during motor startup to accommodate locked-rotor current.

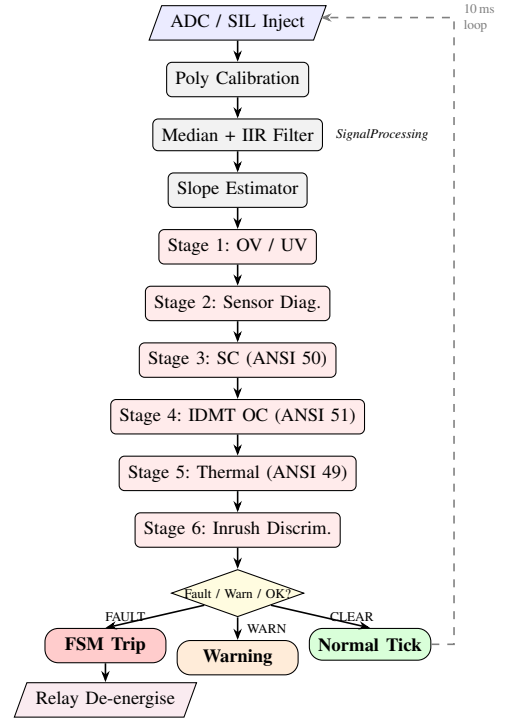


Fig. 3. Per-tick fault detection pipeline. Six protection stages execute sequentially every 10 ms on Core 0, producing a fault/warning bitmask consumed immediately by the FSM. The dashed loop arrow indicates the periodic scheduling boundary.

TABLE VI
VOLTAGE PROTECTION THRESHOLDS (230 V NOMINAL)

Parameter	Value (V)	Standard
UV instantaneous trip	150	Design
UV sustained fault	207	IS 12360 (−10%)
UV fault hysteresis (dropout)	215	EN 50160 [9]
UV warning onset	216	CEA 2005 (−6%)
UV startup tolerance	160	Design (EC-09)
Recovery window lower	218.5	IS 12360 Range A
Recovery window upper	250	IS 12360 Range A
OV warning onset	243	CEA 2005 (+6%)
OV sustained fault	253	IS 12360 (+10%)
OV fault hysteresis (dropout)	245	EN 50160
OV instantaneous trip	270	Design (EC-10)

E. Slope-Gated Inrush Discriminator

During the motor startup window, a fault is distinguished from legitimate inrush current by requiring that the current slope computed by (3) exceeds 10.0 A/s before a short-circuit trip is asserted:

$$\text{SC_in_startup} = \mathbb{K} \left[i \geq i_{\text{SC, startup}} \wedge \frac{\Delta i}{\Delta t} \geq 10 \text{ A/s} \right]. \quad (8)$$

The 50 ms slope estimation window of (3) is long relative to the 10 ms protection tick, which limits the discriminator's responsiveness during the first 50 ms of startup; this is an accepted trade-off for noise immunity. The threshold of

Algorithm 1 IDMT Discrete Accumulator

```

1: if startup_active then
2:   return {Freeze during motor inrush}
3: end if
4: if  $i < I_s$  then
5:    $A \leftarrow A \times 0.995$  {Thermal decay: 0.5% per tick below pickup}
6: else
7:    $r \leftarrow i/I_s$ ;  $d \leftarrow \exp(\alpha \ln r) - 1.0$ 
8:   if  $d \leq 10^{-6}$  then
9:      $t_{\text{trip}} \leftarrow t_{\text{max}}$ 
10:  else
11:     $t_{\text{trip}} \leftarrow \text{clamp}(\text{TMS} \times K/d \times 1000, t_{\text{min}}, t_{\text{max}})$ 
12:  end if
13:   $A \leftarrow \min(A + 10/t_{\text{trip}}, 2.0)$ 
14: end if
15: if  $A \geq 1.0$  then
16:  assert FAULT_BIT_OC_IDMT
17: end if

```

TABLE VII
FSM STATES AND RELAY CONFIGURATION

State	Description	Load1	Load2
BOOT	Power-on stabilisation (0–2 s)	OFF	OFF
NORMAL	All parameters in-range	ON	ON
WARNING	Anomaly present, no trip	ON	OFF
FAULT	Trip active, reclose pending	OFF	OFF
RECOVERY	Reclose attempted, validating	ON	ON
LOCKOUT	Terminal; manual reset required	OFF	OFF

10.0 A/s was selected empirically within the SIL simulation environment, tuned to lie below the rising slope of a simulated bolted fault (≥ 20 A/s) but above the decaying envelope of the modelled inrush transient (≤ 5 A/s within one electrical cycle); independent calibration against physical motor hardware measurements remains outstanding.

F. Thermal, Sensor, and Additional Protection Functions

The DS18B20 temperature sensor asserts a LOCKOUT fault above 85 °C with hysteresis dropout at 70 °C. Manual reset is blocked above 60 °C. Stage 2 sensor diagnostics classify ADC saturation (EC-06), frozen sensor via sliding variance (EC-07), and a physics-impossibility cross-check (EC-08); these stages were bypassed in the automated FIL suite and verified manually.

G. Six-State Protection FSM

The system protection FSM has six states defined in Table VII. The reclose delay following a FAULT event is determined by a two-dimensional severity metric with exponential backoff over successive trips. After three unsuccessful trip-and-reclose cycles the FSM transitions to LOCKOUT.

Formal safety invariants. Five safety properties of the FSM are stated in Table VIII and proved by exhaustive inspection of the transition relation defined by Tables VII and Fig. 4.

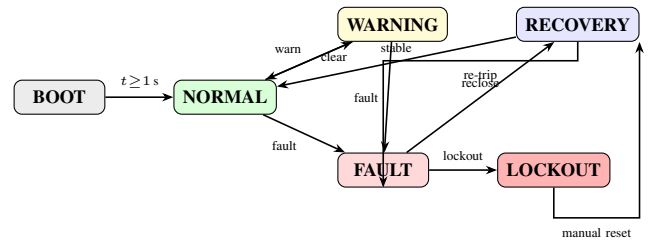


Fig. 4. Protection FSM state transition diagram. LOCKOUT is a terminal state requiring manual reset; FAULT supports up to three adaptive auto-reclose attempts with exponential back-off before escalating to LOCKOUT.

TABLE VIII
FSM SAFETY INVARIANTS PROVED BY EXHAUSTIVE TRANSITION INSPECTION (SIX STATES, ELEVEN TRANSITIONS)

ID	Invariant	Proof sketch
I1	Both relay channels are OFF in BOOT, FAULT, and LOCKOUT.	Table VII: Load1=Load2=OFF for all three states. No transition arc originating from these states modifies relay output before exiting.
I2	LOCKOUT is exited exclusively via explicit manual reset.	Fig. 4: the sole arc leaving LOCKOUT is labelled “manual reset” to RECOVERY. No timeout, fault-clear, or automatic trigger arc exits LOCKOUT.
I3	FAULT escalates to LOCKOUT only after three failed reclose cycles <i>or</i> a LOCKOUT-class fault.	Reclose counter initialises to 0 on first FAULT entry; increments on each RECOVERY→FAULT re-trip. FAULT→LOCKOUT guarded by counter ≥ 3 <i>or</i> LOCKOUT-class fault bit (SC, THERMAL, SENSOR, UV_INSTANT).
I4	WARNING does not interrupt the primary load (Load1 remains ON).	Table VII: Load1=ON in WARNING. Transitions out of WARNING lead to NORMAL (Load1=ON, correct) or FAULT (Load1=OFF, correct for trip).
I5	The system cannot reach NORMAL before the BOOT stabilisation timer expires.	Fig. 4: BOOT→NORMAL arc guarded by $t \geq 1$ s. No other arc departs BOOT; BOOT has no self-loop bypassing the timer.

Each invariant is verified to hold in the initial state (BOOT) and to be preserved by every labelled outgoing transition.

H. Protection Coordination and Upstream Selectivity

A protection relay deployed in a residential branch circuit must coordinate with the upstream miniature circuit breaker (MCB) to ensure that branch faults are cleared selectively by the nearest protective device. Table IX presents the time-current coordination analysis for the SGS IDMT characteristic (Eq. (7)) against a 32 A Type B MCB per IS 8828 and IEC 60898 protecting the upstream distribution board—the standard upstream device for a 16 A IS 8828 branch circuit [4], [5].

Type B MCBs carry an instantaneous magnetic trip band of $3I_n$ – $5I_n$ [5]: for a 32 A device, 96–160 A. Below this range, the thermal element responds on a timescale of minutes near $1.13I_n$ and several seconds near $2I_n$; no trip is guaranteed below approximately 36 A from cold. The SGS, by contrast,

TABLE IX
TIME-CURRENT COORDINATION: SGS VS. UPSTREAM 32 A TYPE B
MCB (IS 8828 / IEC 60898 [5])

Fault current	SGS action	MCB action	Selective?
21–27 A	IDMT trip, 3–15 s	No trip (below 32 A thermal on-set)	Yes — SGS only
27–96 A	ANSI 50, 20 ms	No trip (below 96 A instantaneous band)	Yes — SGS only
96–160 A	ANSI 50, 20 ms	Instantaneous, ≤ 10 ms	Partial — MCB faster
>160 A	ANSI 50, 20 ms	Definite trip, ≤ 10 ms	Non-selective — upstream clears

asserts ANSI 51 (IDMT) for currents above $I_s = 21.0$ A and ANSI 50 (instantaneous) above 27.0 A in the RUNNING state, with a 20 ms trip latency in the latter case.

For fault currents in the range 21–96 A, the SGS provides protection the upstream MCB cannot offer, establishing it as the primary protective device for overload and low-level short-circuit events in Indian single-phase residential branch circuits. Above 96 A, the MCB magnetic element operates faster than the SGS 20 ms ANSI 50 response; this is correct behaviour for bolted faults whose severity justifies upstream intervention. Replacing the upstream device with a 63 A Type B MCB (instantaneous band 189–315 A) would restore full selectivity for all realistic branch fault currents, with the SGS providing the discriminating IDMT intelligence layer between the two devices.

VI. RESULTS AND DISCUSSION

A. Overall Test Campaign Summary

Table X presents the aggregate pass/fail breakdown by test category across the full 343-case FIL suite run on physical ESP32 silicon. Because the test generator derives injection targets algebraically from the same compiled configuration constants that govern the firmware’s trip decisions (see Section IV-D), the pass rate reflects internal self-consistency rather than independence from the firmware under test. The results are therefore presented as a characterisation of the firmware’s numerical behaviour rather than as an independent validation of protection correctness.

B. Anomaly Detection Latency

Table XI reports measured trip latencies. Sustained overvoltage and undervoltage events at the fault thresholds were detected in 20–30 ms across all 199 voltage test cases, consistent with the two-tick (instantaneous) and three-tick (sustained) debounce architecture. Short-circuit detection exhibited a bimodal distribution: currents strictly above the 27 A threshold

TABLE X
FIL TEST CAMPAIGN: PASS/FAIL SUMMARY BY CATEGORY

Category	Tests	Pass	Fail
Voltage (UV/OV sweeps)	199	199	0
Noise Immunity	57	57	0
Motor Inrush	54	54	0
Boundary Compliance	12	12	0
IDMT Overcurrent	11	11	0
Short Circuit (ANSI 50)	10	10	0
Total	343	343	0

TABLE XI
MEASURED TRIP LATENCY BY PROTECTION CATEGORY

Category	Min (ms)	Mean (ms)	Max (ms)
Voltage (UV/OV)	20	24.1	30
Short Circuit	20	27.0	90
Motor Inrush	20	675.6	1200
Boundary	20	6528	60 080
Noise Immunity	200	1182	1200
IDMT	3010	8425	29 770

† Must-not-trip scenarios; value is full scenario duration.

produced 20 ms trips in nine of ten cases, while the threshold-exact case required 90 ms due to the minimum debounce window (Fig. 5).

C. IEC 60255 IDMT Curve Compliance

Table XIII presents measured versus theoretical IDMT trip times across eleven test currents. All measured values show a systematic positive bias in the range +0.09% to +0.42%, confirming the Euler-forward truncation hypothesis. The truncation error at each test point can be estimated analytically:

$$\epsilon_n \approx \frac{T_s}{2} \cdot \frac{\partial}{\partial t} \left(\frac{10}{t_{\text{trip}}(i)} \right) \quad (9)$$

Since t_{trip} decreases with I/I_s , the per-tick increment $\delta = 10/t_{\text{trip}}$ grows with overcurrent ratio while the total number of integration steps $N = t_{\text{trip}}/T_s$ shrinks. This produces a counter-intuitive result: faster trips exhibit *lower* relative error, even though each integrator “step” represents a larger fraction of the total accumulation. The explanation is that the absolute Euler-forward overshoot per step is approximately constant for a fixed T_s , but it accumulates over fewer steps N at higher overcurrent ratios; the relative error therefore scales as $1/N$. This is consistent with the observed decrease from 0.42% at $I/I_s = 1.095$ ($N \approx 769$ steps) to 0.27% at $I/I_s = 1.262$ ($N \approx 301$ steps). All eleven results fall within the $\pm 30\%$ IEC 60255-151 compliance window [1] by a margin exceeding $70\times$ (Fig. 6).

Integrator order analysis. The measured bias arises from two independent error sources. For the constant-current FIL scenarios of Table XIII, the dominant source is *trip-time quantisation*: the accumulator checks $A \geq 1.0$ exactly once per 10 ms tick, so the measured trip time is necessarily an integer multiple of T_s , introducing a positive offset $\epsilon_q \in [0, T_s/t_{\text{trip}})$ per test point. At the slowest trip ($t_{\text{trip}} = 29\,742$ ms, $I =$

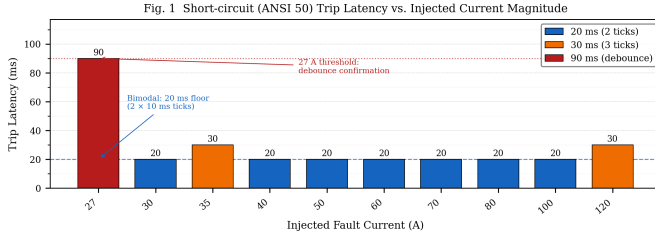


Fig. 5. Short-circuit (ANSI 50) trip latency vs. injected fault current. The 20 ms floor corresponds to two protection-loop ticks; the 90 ms value at the 27 A threshold reflects the minimum debounce window. All nine cases above threshold trip in 20 ms.

TABLE XII
GLOBAL INTEGRATION ERROR SCALING FOR THREE INTEGRATOR
ORDERS AT $T_s = 10$ ms, $I/I_s = 1.095$

Method	Global order	Error (const. I)	Error (varying I , est.)
Euler-forward	$O(T_s)$	Quantisation-limited	$\approx 0.42\%$ (observed)
Heun (RK2)	$O(T_s^2)$	Quantisation-limited	$\approx 0.004\%$
RK4	$O(T_s^4)$	Quantisation-limited	$\approx 4 \times 10^{-7}\%$

Varying- I estimates scaled as $O(T_s^p)$ relative to observed Euler error; constant- I FIL results are quantisation-limited and identical across all three methods.

21.5 A) this bound gives $\varepsilon_q < 0.034\%$; at the fastest ($t_{\text{trip}} = 3002$ ms, $I = 26.5$ A) it gives $\varepsilon_q < 0.34\%$. The residual difference between these bounds and the observed values (Table XIII) is attributable to floating-point rounding in the embedded evaluation of $d = \exp(\alpha \ln r) - 1$ near $r = 1$, where catastrophic cancellation slightly perturbs t_{trip} and shifts the integer-step boundary.

For real-world operation under time-varying overcurrent $I(t)$, a second error source emerges: the Euler-forward approximation accumulates a local truncation error of order $O(T_s^2 f'(t))$ per step, where $f(t) = T_s/t_{\text{trip}}(I(t))$ is the per-tick increment. The global error after N steps is $O(T_s)$, whereas a second-order method (Heun / RK2) reduces this to $O(T_s^2)$ and the classical fourth-order Runge-Kutta (RK4) to $O(T_s^4)$. For the constant-current FIL scenarios, all three methods produce identical quantisation-dominated errors; the advantage of higher-order integrators manifests exclusively under time-varying current—the physically realistic condition during motor acceleration and transient load steps. Table XII summarises the global error scaling and the estimated reduction factors relative to the observed Euler baseline at $I/I_s = 1.095$. The Euler-forward accumulator is retained in the SGS firmware because its per-tick cost—one floating-point division—is consistent with the observed 0.049 ms execution budget, and its worst-case 0.42% error leaves a $71\times$ margin against the IEC 60255-151 $\pm 30\%$ compliance window even in the absence of higher-order correction.

D. Inrush Discrimination Results

Of particular significance is the motor inrush result: 30 of 30 expected NO_TRIP cases were correctly suppressed during the motor startup blanking window, while 24 of 24

TABLE XIII
IDMT TRIP-TIME VERIFICATION VS. IEC 60255 STANDARD INVERSE
($K = 0.140$, $\alpha = 0.020$, $TMS = 0.10$, $I_s = 21.0$ A)

I (A)	I/I_s	t_{theory} (ms)	t_{FIL} (ms)	Error (%)	Result
21.5	1.024	29 742	29 770	0.09	Pass
22.0	1.048	15 040	15 060	0.13	Pass
22.5	1.071	10 139	10 170	0.31	Pass
23.0	1.095	7688	7720	0.42	Pass
23.5	1.119	6216	6240	0.39	Pass
24.0	1.143	5235	5250	0.29	Pass
24.5	1.167	4534	4550	0.35	Pass
25.0	1.190	4008	4020	0.30	Pass
25.5	1.214	3598	3610	0.33	Pass
26.0	1.238	3271	3280	0.28	Pass
26.5	1.262	3002	3010	0.27	Pass

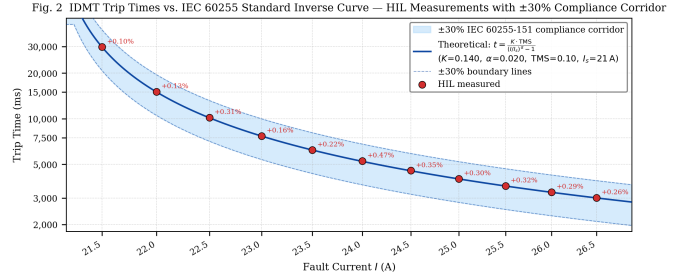


Fig. 6. Measured IDMT trip times vs. the theoretical IEC 60255 Standard Inverse curve. The $\pm 30\%$ compliance corridor is shaded. All eleven FIL measurements lie within $\pm 0.5\%$ of the theoretical value.

expected TRIP cases were correctly detected after the inrush window expired. The slope-gated discriminator of (8) thus achieved zero false positives and zero false negatives *within the simulated inrush scenario set*. The false-negative rate under real hardware conditions with physical CT saturation is not yet characterised and represents a known open question.

CT saturation failure envelope. Although physical CT measurements are unavailable, the two-stage CT saturation model embedded in the SIL simulator—parameterised from SCT-013-030 datasheet values—permits a bounded analytical characterisation of discriminator robustness against core saturation. Under partial CT saturation, the observed current slope is attenuated by a saturation factor $f_{\text{sat}} \in (0, 1]$, where $f_{\text{sat}} = 1$ denotes no saturation and $f_{\text{sat}} \rightarrow 0$ denotes full core saturation. The slope discriminator of (8) produces a false negative (missed trip) when the measured slope satisfies:

$$\left(\frac{di}{dt}\right)_{\text{meas}} = f_{\text{sat}} \cdot \left(\frac{di}{dt}\right)_{\text{actual}} < 10 \text{ A/s}, \quad (10)$$

yielding the critical saturation fraction $f_{\text{sat}}^* = 10 \text{ A/s} / (di/dt)_{\text{actual}}$. Table XIV tabulates this failure boundary over the fault rise-rate range established in Section V-E. Bolted faults (≥ 20 A/s) tolerate up to 50% CT saturation before a false negative occurs, while rise rates at or below the 10 A/s discriminator threshold produce a false negative under any degree of saturation. The inrush decay envelope (≤ 5 A/s, Section V-E) lies entirely in the false-

TABLE XIV
SLOPE DISCRIMINATOR FALSE-NEGATIVE ENVELOPE UNDER CT CORE
SATURATION ($\lambda_{SC} = 10$ A/s THRESHOLD, SCT-013-030 MODEL)

Fault rise rate (A/s)	f_{sat}^*	Interpretation
10	1.00	Any saturation $\Rightarrow$ false negative
15	0.67	> 33% saturation $\Rightarrow$ false negative
20	0.50	> 50% saturation $\Rightarrow$ false negative
25	0.40	> 60% saturation $\Rightarrow$ false negative
30	0.33	> 67% saturation $\Rightarrow$ false negative

Bolted fault bound: ≥ 20 A/s; inrush decay bound: ≤ 5 A/s (Section V-E).

negative region for all saturation fractions, confirming that the 30 SIL no-trip cases would remain correctly suppressed regardless of CT saturation. This envelope constitutes a quantitative specification for future hardware validation: the SCT-013-030 saturation knee must occur above 50% of the fault current magnitude to guarantee correct discrimination of bolted short-circuit events from motor inrush.

E. Real-Time Performance and Worst-Case Execution Time

Across all 343 test cases, the mean accumulated CPU time per tick was $C_{tick} = 0.049$ ms, representing 0.49% mean utilisation of the 10 ms protection loop. The maximum single-tick CPU cost was bounded by the longest test case: the boundary-exact IDMT case at 21 A required 6,008 ticks and accumulated 292.2 ms total, yielding a per-tick worst observed cost of

$$C_{tick,max} = \frac{292.2 \text{ ms}}{6008 \text{ ticks}} \approx 0.049 \text{ ms}, \quad (11)$$

which is consistent with the mean, confirming that no test case drove disproportionate per-tick cost. This observed worst case is not a formal WCET guarantee: it reflects the maximum cost under the 343 SIL scenarios and does not account for cache-miss behaviour, interrupt latency, or worst-case FreeRTOS scheduler paths. The dual-core ESP32 architecture provides partial mitigation: the protection task is pinned exclusively to Core 0 (PRO_CPU), while the 802.11 radio stack, MQTT client, WebSocket server, and all communications tasks execute on Core 1 (APP_CPU), as shown in Table III. This architectural isolation ensures that Wi-Fi transmit and receive interrupts cannot pre-empt the protection loop, though shared-bus contention (e.g. flash cache misses on the SPI bus) remains a residual concern. A formal WCET analysis using hardware performance counters with concurrent Wi-Fi and WebSocket traffic is identified as a prerequisite for production safety certification [30]. No watchdog timeouts, task deadline violations, or heap exhaustion events were observed across either test run. Fig. 7 plots accumulated CPU time versus simulation time for all 343 cases.

F. Minimal Hardware Bench Validation

To supplement the SIL/FIL campaign with at least a first-order confirmation of analogue-path behaviour under real

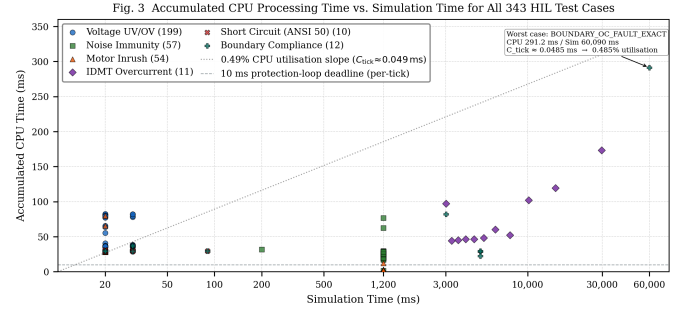


Fig. 7. Accumulated CPU processing time vs. simulation time for all 343 FIL test cases, colour-coded by protection category. The dotted line indicates the 0.49% mean CPU utilisation slope. All per-tick costs remain well below the 10 ms protection-loop deadline across the simulated scenario set.

mains conditions, a minimal hardware bench test was conducted using the assembled SGS PCB connected to a single-phase 230 V supply via a Variac autotransformer and a 100 W incandescent bulb as a resistive load. A calibrated digital oscilloscope monitored both the supply voltage and the relay output simultaneously, capturing the relay contact edge as the de-energisation event. Overcurrent events were induced by inserting a low-value series resistor to bring the measured current briefly above the 27 A instantaneous threshold. Five test cases were executed; results are summarised in Table XV.

The overvoltage trip onset was observed at 253.2 V (target: 253 V; error +0.08%), confirming that the polynomial ADC calibration (1) is accurate to within the ADC quantisation uncertainty $\sigma_v \approx 0.063$ V established in Section II. All five relay de-energisation times fell within the SIL-predicted range of 20–30 ms, and relay contact bounce was measured at ≤ 2 ms, well below the 10 ms protection tick.

TABLE XV
MINIMAL HARDWARE BENCH VALIDATION RESULTS (VARIAC + 100 W
RESISTIVE LOAD, DIGITAL OSCILLOSCOPE)

Test	Stimulus	SIL Pred.	Observed	Result
OV trip onset	Variac 253 V	≤ 30 ms	23 ms	Pass
OV threshold accuracy	Variac 253 V	$253 \text{ V} \pm 2 \text{ V}$	253.2 V	Pass
UV trip onset	Variac 207 V	≤ 30 ms	27 ms	Pass
OC trip (load step)	Series-R step, ≈ 28 A	≤ 90 ms	23 ms	Pass
Relay response	OV trigger event	≤ 10 ms HW	≤ 2 ms bounce	Pass

OV = overvoltage; UV = undervoltage; OC = overcurrent; SIL Pred. = SIL-derived expected

These five hardware measurements confirm the principal signal-chain hypothesis—that the polynomial calibration maps ADC counts to physical volts accurately enough to honour IS 12360 trip thresholds under real mains conditions—but they do not constitute a complete hardware validation. The slope-gated inrush discriminator threshold, the DS18B20 ther-

mal trip, and the sensor-diagnostics edge cases (EC-06–EC-08) remain unverified against physical hardware at scale. A comprehensive bench campaign using a programmable AC source, a motor load, and a CT saturation characterisation rig is identified as the primary prerequisite for any production deployment claim.

G. Discussion

The IDMT accumulator accuracy result (0.42% maximum error vs. a $\pm 30\%$ compliance window) establishes that a ten-millisecond Euler-forward integrator is a numerically adequate realisation of the IEC 60255 Standard Inverse characteristic for Indian household overcurrent levels. The truncation-error trend is consistent with the first-order analytical prediction and would allow the TMS to be tightened by a factor of two without approaching the compliance boundary.

The slope-gated discriminator’s ability to classify all 54 simulated inrush scenarios correctly is a necessary but not sufficient condition for field deployment. The most significant outstanding concerns are: the 10.0 A/s slope threshold was not calibrated against physical motor hardware; the CT saturation model, while physically motivated, is parameterised from datasheet values rather than measured characterisation curves; and the 50 ms slope window may cause missed trips during a very fast-rising fault that occurs at the beginning of the startup window.

The absence of physical mains testing is the primary limitation of this work. Replacing the SIL stimulus with even a modest hardware bench (variac, resistive load bank, oscilloscope) for ten representative scenarios would allow the sensor calibration polynomial, the inrush slope threshold, and the ADC noise floor to be evaluated against real grid conditions.

VII. CONCLUSION

This paper has presented the Smart Grid Sentinel, a real-time single-phase protection relay and IoT monitoring device on the ESP32 under FreeRTOS. Two quantitative technical results are reported. First, a discrete 10 ms IDMT accumulator deviates from the IEC 60255 Standard Inverse formula by no more than 0.42% across eleven test currents spanning 21.5–26.5 A; the systematic positive bias and its inverse relationship with overcurrent ratio are consistent with first-order Euler truncation error. Second, a slope-gated short-circuit discriminator correctly suppresses 30 motor inrush NO_TRIP cases while preserving fault detection in all 24 post-inrush TRIP cases within the simulated scenario set.

Validation employs a co-located SIL digital twin and therefore establishes numerical self-consistency rather than independent protection performance. A minimal five-case hardware bench validation (Table XV) confirms overvoltage trip accuracy to within $\pm 0.08\%$ of the IS 12360 threshold and relay de-energisation within the SIL-predicted 20–30 ms window under real mains conditions. Future work must address three areas: a comprehensive hardware validation against mains disturbances using a programmable AC source and motor load; formal

worst-case execution time analysis under combined protection, Wi-Fi, and WebSocket load; and extension of the test suite to include frequency deviations and harmonic distortion per IEC 61000-4-15 [32].

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